Prompt: Write a recursive, object-oriented C++ program that solves a Sudoku puzzle on a 9x9 grid subdivided into 9 3x3 subgrids. Following that, if given a partially complete puzzle, it can find a possible solution if one exists or prove that no solution exists. Outputs expected are: puzzle file name, puzzle from the file printed out, solved puzzle printed out, or "No solution found" if it is proven not to be solvable through exhaustive means; if not unique solution, print all possible solutions. When creating solution.txt files, they should follow the naming convention: solution1.txt is tied to puzzle1.txt

GenAI used: I used 365 Copilot through from their desktop app and Claude from the web browser

Justification: I chose to use the claude generated code this time due to it using a bitmasking which I wanted to work more with since I have done that less I used both stack overflow and geeks for geeks to get more understanding of how to work with this as I went through and looked for what could be improved in the files. The time and space complexity were better with the claude version also making it a better choice to start with outside of my personal reason. The long term maintainability of the claude code was also generally more maintainable since it was decomposed into more classes that make it easier in the long run. I made a small improvement to the maintainability in adding the SolutionListenr and SolutionPrinter classes that will write out the outputs.